

Tuberculosis Detection Using Chest X-ray Images

Department of Computer Science and Engineering
Kongu Engineering College
Tamil Nadu, India

Abstract—Tuberculosis (TB) continues to be one of the major respiratory diseases affecting millions of people worldwide. Rapid and accurate identification of TB is essential for early treatment and preventing disease transmission. This study presents an automated framework for tuberculosis detection from chest X-ray images using an Enhanced CoAtNet deep learning model. Chest X-ray images collected from the Montgomery and Shenzhen public datasets are combined to create a diverse dataset. Before classification, the images undergo preprocessing steps including Contrast Limited Adaptive Histogram Equalization (CLAHE), resizing, normalization, and lung segmentation to improve image quality and focus on the lung region. The processed images are then classified into tuberculosis and normal categories using the Enhanced CoAtNet architecture. To increase the transparency of the prediction process, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to identify the image regions that influence the model's decision. The proposed framework aims to provide an accurate, reliable, and interpretable computer-aided diagnostic system that can assist healthcare professionals during TB screening.

Index Terms—Tuberculosis, Chest X-ray, CoAtNet, Deep Learning, Grad-CAM, Lung Segmentation

I. INTRODUCTION

Tuberculosis (TB) is a contagious infectious disease caused by *Mycobacterium tuberculosis* that mainly affects the lungs. It remains one of the leading causes of death worldwide, particularly in developing countries where access to advanced healthcare facilities is limited. Early detection of tuberculosis is essential to reduce disease transmission, improve patient survival, and ensure timely treatment. Among the various diagnostic techniques, Chest X-ray (CXR) imaging is widely used as a primary screening tool because it is non-invasive, cost-effective, and easily available. However, manual interpretation of chest X-ray images depends on the experience of radiologists and may lead to delayed or inaccurate diagnosis due to variations in image quality and subtle disease patterns.

Recent developments in Artificial Intelligence (AI) and Deep Learning (DL) have significantly improved the automatic analysis of medical images. Deep learning models can automatically learn important image features and provide reliable disease classification without manual feature extraction. Conventional Convolutional Neural Networks (CNNs) are effective in capturing local image features, whereas Transformer-based models are capable of learning global contextual information. Hybrid architectures combine these advantages to achieve better classification performance.

In this research, an Enhanced CoAtNet model is proposed for automated tuberculosis detection using chest X-ray images. CoAtNet combines convolutional operations with transformer

attention mechanisms to effectively learn both local and global image representations. This hybrid architecture improves feature extraction and enhances the accuracy of tuberculosis classification.

The proposed framework utilizes two publicly available chest X-ray datasets: the Montgomery County dataset and the Shenzhen dataset. The Montgomery dataset contains 138 chest X-ray images, including 80 normal and 58 tuberculosis cases. The Shenzhen dataset consists of 662 images, including 326 normal and 336 tuberculosis cases. After integrating both datasets, a total of 800 chest X-ray images are obtained, comprising 406 normal and 394 tuberculosis images. The combined dataset provides greater diversity in patient characteristics and imaging conditions, helping the model achieve better robustness and generalization.

Before classification, all chest X-ray images undergo preprocessing to improve image quality and reduce unwanted variations. Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to enhance image contrast, followed by image resizing and normalization to maintain a consistent input format. Lung segmentation is then performed to isolate the lung region and remove irrelevant background information, enabling the model to focus only on the area of clinical importance.

The preprocessed images are classified into tuberculosis and normal classes using the Enhanced CoAtNet model. To improve the transparency and interpretability of the proposed system, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated as an Explainable Artificial Intelligence (XAI) technique. Grad-CAM generates heatmaps that highlight the regions of the chest X-ray image contributing to the model's prediction. These visual explanations help clinicians understand the decision-making process and increase confidence in AI-assisted diagnosis.

The proposed framework integrates dataset fusion, image preprocessing, lung segmentation, Enhanced CoAtNet classification, and Grad-CAM visualization to develop an accurate and explainable computer-aided diagnosis system for tuberculosis detection. The system is expected to support healthcare professionals by providing reliable predictions and improving the efficiency of TB screening in clinical practice.

II. LITERATURE SURVEY

A. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) have been widely used for tuberculosis detection because of their ability to extract local image features from chest X-ray images. Popular

CNN models such as VGG19, ResNet, EfficientNet, and Xception have achieved high classification accuracy. However, CNN models mainly focus on local information and often fail to capture global contextual relationships within medical images.

B. Vision Transformers

Vision Transformers (ViTs) utilize self-attention mechanisms to learn global features from chest X-ray images. They have demonstrated improved performance in several medical imaging tasks. However, Vision Transformers require large datasets and high computational resources, making them less suitable for small medical datasets.

C. Hybrid Deep Learning Models

Hybrid architectures combine the advantages of CNNs and Vision Transformers. CoAtNet is one such hybrid model that integrates convolution operations for local feature extraction and transformer attention for global context learning. This combination improves feature representation and classification accuracy for chest X-ray image analysis.

D. Research Gap

Existing tuberculosis detection methods mainly rely on CNNs, Vision Transformers, or hybrid deep learning models. Although these approaches achieve promising classification performance, several challenges remain. Many studies use limited datasets, reducing the model's ability to generalize across diverse patient populations.

In addition, some existing frameworks do not incorporate effective image preprocessing and lung segmentation, which can affect feature extraction and overall classification accuracy. Furthermore, many deep learning models function as black-box systems, making it difficult for clinicians to understand the reasoning behind their predictions.

To address these limitations, this work proposes an Enhanced CoAtNet-based framework that integrates the Montgomery and Shenzhen chest X-ray datasets, applies CLAHE, resizing, normalization, and lung segmentation, and employs Grad-CAM for explainable prediction. The proposed framework aims to improve classification accuracy while providing transparent and reliable decision support for tuberculosis diagnosis.

““latex

III. METHODOLOGY

The proposed methodology consists of several stages including dataset collection, image preprocessing, lung segmentation, label preparation, dataset splitting, data augmentation, model development, model training, model evaluation, explainable AI, automated report generation, and web application integration.

A. Dataset Collection

The system uses two publicly available chest X-ray datasets: the Shenzhen Chest X-ray Dataset and the Montgomery Chest X-ray Dataset. The combined dataset contains 800 chest X-ray images, including 406 normal images and 394 tuberculosis images.

B. Image Preprocessing

The collected chest X-ray images are preprocessed before model training. The images are converted into a consistent format and resized to 224×224 pixels. Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to improve image contrast. Noise and unnecessary variations are also reduced to improve the quality of the input images.

C. Lung Segmentation

A pretrained TorchXRyVision PSPNet model is used to identify and extract the lung regions from the chest X-ray images. This process removes unnecessary background regions and allows the classification model to focus mainly on the lung area where tuberculosis-related abnormalities may occur.

D. Label Preparation

Each image is assigned one of two class labels: *Normal* or *Tuberculosis*. These labels are converted into numerical values suitable for deep learning model training.

E. Dataset Splitting

The processed images are divided into training, validation, and testing datasets. The training dataset is used to train the model, the validation dataset is used to monitor and tune model performance, and the testing dataset is used to evaluate the final performance of the trained model.

F. Data Augmentation

Data augmentation is applied to improve model generalization and reduce overfitting. Rotation, horizontal flipping, zooming, and brightness adjustment are applied to the training images. These transformations create variations of the original images while maintaining their original class labels.

G. Model Development

The proposed system uses an Enhanced CoAtNet deep learning model. CoAtNet combines Convolutional Neural Network (CNN) operations with Transformer-based attention mechanisms. CNN layers are used to extract local features such as edges, textures, and lung patterns, while Transformer layers learn global relationships within the X-ray image. The model classifies the input image as either Tuberculosis or Normal.

H. Model Training

The Enhanced CoAtNet model is trained using the preprocessed and augmented chest X-ray images. During training, the model predicts the class of an input image and compares the prediction with the actual label. The error is calculated using a loss function. Backpropagation is then used to calculate gradients and update the model weights using an optimizer. Training continues until stable validation performance is achieved.

I. Model Evaluation

The trained model is evaluated using Accuracy, Precision, Recall, F1-score, and Area Under the Curve (AUC). The obtained performance is shown below.

TABLE I
PERFORMANCE OF THE PROPOSED MODEL

Metric	Result
Validation Accuracy	85.83%
Test Accuracy	77.50%
Precision	83.33%
Recall	67.80%
F1-score	74.77%
AUC	0.8852

J. Explainable Artificial Intelligence

Grad-CAM is used to improve the interpretability of the model. It generates a heatmap highlighting the regions of the chest X-ray image that influenced the model's prediction. This helps users understand whether the model focused on relevant lung regions during classification.

K. Automated Report Generation

For every uploaded X-ray image, the system generates an automated report containing the predicted result, confidence score, Grad-CAM visualization, clinical interpretation, and medical disclaimer.

L. Web Application Integration

The trained model is integrated with a web application. The user can create an account or log in, enter patient details, and upload a chest X-ray image. The system processes the image and displays the predicted result and confidence score. The Grad-CAM visualization is also displayed to show the important regions considered by the model. The generated diagnostic report can be reviewed or downloaded.

...

TABLE II
BASE PAPER COMPARISON

S.No	Paper Title	Methodology	Performance	Advantages	Limitations
1	Deep Learning-Based Automatic Detection of Tuberculosis Disease in Chest X-Ray Images (2022)	Transfer learning using VGG16, VGG19, ResNet50, Xception and InceptionV3.	90% Accuracy	Improves TB detection accuracy and reduces training time.	Small dataset and no Explainable AI.
2	TB-CXRNet: Tuberculosis and Drug-Resistant Tuberculosis Detection Technique Using Chest X-ray Images (2024)	Developed a CNN-based deep learning framework (TB-CXRNet) with Score-CAM for detecting tuberculosis and drug-resistant tuberculosis from chest X-ray images.	93.32% Accuracy	Supports both TB and drug-resistant TB detection while improving model interpretability using Score-CAM.	Focuses only on CNN architecture and lacks Transformer-based feature learning.
3	Ensemble Technique Coupled with Deep Transfer Learning for Tuberculosis Detection (2022)	U-Net segmentation with ensemble CNN models.	98% Accuracy	Better lung segmentation and robust prediction.	No Explainable AI.
6	Detecting Tuberculosis-Consistent Findings in Lateral Chest X-Rays Using an Ensemble of CNNs and Vision Transformers (2024)	An ensemble framework combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to identify tuberculosis-consistent findings from lateral chest X-ray images.	95.7% Accuracy	Combines local and global feature extraction, improving detection performance and robustness for lateral chest X-ray analysis.	High computational complexity and does not include Explainable AI techniques such as Grad-CAM.
5	Deep Learning Architecture for Multi-Class Lung Disease Classification (2023)	CNN model for TB, Pneumonia, COVID-19 and Normal images.	96% Accuracy	Supports multi-class disease detection.	Limited interpretability.
6	Deep Learning Classification of Active Tuberculosis Lung Zone-Wise Manifestations Using Chest X-Ray Images (2023)	CNN-based deep learning model for classifying active tuberculosis by analyzing lung zone-wise manifestations in chest X-ray images.	96.8% Accuracy	Provides lung zone-wise localization of TB abnormalities and improves disease interpretation.	Does not use Transformer-based architectures or Explainable AI techniques such as Grad-CAM.
7	TB-CXRNet: Tuberculosis and Drug-Resistant Tuberculosis Detection Using Chest X-Ray Images (2024)	CNN with Score-CAM explainability.	93.32% Accuracy	Supports MDR-TB detection and visualization.	CNN only; no Transformer.
8	Tuberculosis Detection from Chest X-Ray Image Modalities Based on Transformer and CNN (2024)	Hybrid DeiT Transformer with ResNet16.	99.38% Accuracy	Combines local and global feature extraction.	No lung segmentation.
9	Enhanced CoAtNet Based Hybrid Deep Learning Architecture for Automated Tuberculosis Detection in Human Chest X-Rays (2025)	Enhanced CoAtNet with CLAHE pre-processing and LIME explainability.	86.39% Accuracy	Hybrid CNN-Transformer with Explainable AI.	No lung segmentation; moderate accuracy.
10	Ensemble Deep Learning Architectures for Tuberculosis Detection (2026)	Ensemble CNN models with feature fusion and multi-scale learning.	AUC = 0.98	High robustness and generalization.	High computational cost.